

Types	$\tau ::= \text{bool} \mid \tau \rightarrow \tau$
Expressions	$e ::= x \mid \bar{b} \mid \lambda x : \tau. e \mid e e$
Values	$v ::= \bar{b} \mid \lambda x : \tau. e$
Contexts	$\Gamma ::= \emptyset \mid \Gamma, x : \tau$
Substitutions	$\sigma ::= \text{id} \mid \sigma[x \mapsto v]$

Fig. 1. Syntax

$\frac{\text{Typ-Var}}{x : \tau \in \Gamma}$	$\frac{\text{Typ-Blit}}{\Gamma \vdash \bar{b} : \text{bool}}$	$\frac{\text{Typ-Abs}}{\Gamma, x : \tau_1 \vdash e : \tau_2}$	$\frac{\text{Typ-App}}{\Gamma \vdash e_1 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash e_2 : \tau_1}$
$\Gamma \vdash x : \tau$		$\Gamma \vdash \lambda x : \tau_1. e : \tau_1 \rightarrow \tau_2$	$\Gamma \vdash e_1 e_2 : \tau_2$

Fig. 2. Syntactic Typing

$\frac{\text{Red-Beta}}{(\lambda x : \tau. e) v \hookrightarrow e[v/x]}$	$\frac{\text{Red-App}_1}{e_1 \hookrightarrow e'_1 \quad e_1 e_2 \hookrightarrow e'_1 e_2}$	$\frac{\text{Red-App}_2}{e_2 \hookrightarrow e'_2 \quad v e_2 \hookrightarrow v e'_2}$
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Fig. 3. Reduction

$$\begin{aligned} \text{id}(e) &:= e \\ \sigma[x \mapsto v](e) &:= \sigma(e[v/x]) \end{aligned}$$

Fig. 4. Substitution

$$\begin{aligned} \mathcal{V}[\![\text{bool}]\!] &:= \{\bar{b} \mid b \in \mathbb{B}\} \\ \mathcal{V}[\![\tau_1 \rightarrow \tau_2]\!] &:= \{\lambda x : \tau_1. e \mid \forall v \in \mathcal{V}[\![\tau_1]\!]. e[v/x] \in \mathcal{E}[\![\tau_2]\!]\} \\ \mathcal{E}[\![\tau]\!] &:= \{e \mid \exists v. e \hookrightarrow^* v \wedge v \in \mathcal{V}[\![\tau]\!]\} \\ \mathcal{G}[\![\emptyset]\!] &:= \{\text{id}\} \\ \mathcal{G}[\![\Gamma, x : \tau]\!] &:= \{\sigma[x \mapsto v] \mid \sigma \in \mathcal{G}[\![\Gamma]\!] \wedge v \in \mathcal{V}[\![\tau]\!]\} \end{aligned}$$

Fig. 5. Logical Relation

$$\Gamma \models e : \tau := \forall \sigma \in \mathcal{G}[\![\Gamma]\!]. \sigma(e) \in \mathcal{E}[\![\tau]\!]$$

Fig. 6. Semantic Typing